

Analysis on the Real Line Hw 3

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BS - Page 60 Exercise 16

Show that $\lim_{n \rightarrow \infty} \frac{2^n}{n!} = 0$

[Hint: if $n \geq 3$, then $0 < \frac{2^n}{n!} \leq 2(\frac{2}{3})^{n-2}$]

Claim: $2(\frac{2}{3})^{n-2}$ converges to 0, i.e. $\forall \epsilon > 0 \exists N \in \mathbb{N}$ s.t. $|2(\frac{2}{3})^{n-2}| < \epsilon \forall n \geq N$

$$\begin{aligned} 2 \left(\frac{2}{3} \right)^{n-2} &< \epsilon \\ (n-2) \ln\left(\frac{2}{3}\right) &< \ln\left(\frac{\epsilon}{2}\right) \\ n &> \frac{\ln\left(\frac{\epsilon}{2}\right)}{\ln\left(\frac{2}{3}\right)} + 2 \end{aligned}$$

Thus for $N > \max\{3, \frac{\ln(\frac{\epsilon}{2})}{\ln(\frac{2}{3})} + 2\}$ the inequality holds and hence $2(\frac{2}{3})^{n-2}$ converges to 0

Then from the hint, we have $0 < \frac{2^n}{n!} \leq 2(\frac{2}{3})^{n-2} \quad \forall n \geq 3$

Both $\{0\}$ and $\{2(\frac{2}{3})^{n-2}\}$ converge to 0, thus by Squeeze Theorem $\lim_{n \rightarrow \infty} \frac{2^n}{n!} = 0$

BS - Page 67 Exercise 9

Let $y_n := \sqrt{n+1} - \sqrt{n}$ for $n \in \mathbb{N}$. Show that (y_n) and $(\sqrt{n}y_n)$ converge. Find their limits

(a)

We can rewrite $y_n = \sqrt{n+1} - \sqrt{n} = \frac{(\sqrt{n+1}-\sqrt{n})(\sqrt{n+1}+\sqrt{n})}{\sqrt{n+1}+\sqrt{n}} = \frac{1}{\sqrt{n+1}+\sqrt{n}}$ which creates a nicer form to work with.

We have $\sqrt{n+1} + \sqrt{n} > 0 \Rightarrow \frac{1}{\sqrt{n+1}+\sqrt{n}} > 0$

Also $\sqrt{n+1} + \sqrt{n} > 2\sqrt{n} \Rightarrow \frac{1}{2\sqrt{n}} > \frac{1}{\sqrt{n+1}+\sqrt{n}}$

Thus $0 < \{y_n\} < \{\frac{1}{2\sqrt{n}}\} \quad \forall n$

Claim: $\frac{1}{2\sqrt{n}}$ converges to 0, i.e. $\forall \epsilon > 0 \exists N \in \mathbb{N}$ s.t. $|\frac{1}{2\sqrt{n}}| < \epsilon \forall n \geq N$

$$\begin{aligned} \left| \frac{1}{2\sqrt{n}} \right| &< \epsilon \\ \frac{1}{2\sqrt{n}} &< \epsilon \\ n &> \frac{1}{4\epsilon^2} \end{aligned}$$

Thus for $N > \frac{1}{4\epsilon^2}$ the inequality holds and hence $\{\frac{1}{2\sqrt{n}}\}$ converges to 0

Since the constant sequence $\{0\}$ and $\{\frac{1}{2\sqrt{n}}\}$ both converge to 0, we also have that $\{y_n\}$ converges to 0, by the squeeze theorem.

(b)

By dividing top and bottom by $\sqrt{n}$, we can rewrite $\{\sqrt{n}y_n\}$ as

$$\frac{\sqrt{n}}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{1 + \frac{1}{n}} + 1}$$

By using the laws for limits established in class, we can see that

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} &= \frac{\lim_{n \rightarrow \infty} 1}{\lim_{n \rightarrow \infty} \sqrt{1 + \frac{1}{n}} + 1} \\ &= \frac{1}{(\lim_{n \rightarrow \infty} \sqrt{1 + \frac{1}{n}}) + 1}\end{aligned}$$

Claim: $\lim_{n \rightarrow \infty} \sqrt{1 + \frac{1}{n}}$ converges to 1, i.e. $\forall \epsilon > 0 \exists N \in \mathbb{N}$ s.t. $|\sqrt{1 + \frac{1}{n}} - 1| < \epsilon \forall n \geq N$

$$\begin{aligned}\sqrt{1 + \frac{1}{n}} - 1 &< \epsilon \\ 1 + \frac{1}{n} &< \epsilon^2 \\ \frac{1}{\epsilon^2 - 1} &< n\end{aligned}$$

Thus for $N > \frac{1}{\epsilon^2 - 1}$ the inequality holds and hence $\{\sqrt{1 + \frac{1}{n}}\}$ converges to 1

Then,

$$\lim_{n \rightarrow \infty} \sqrt{n}y_n = \frac{1}{1 + 1} = \frac{1}{2}$$

BS - Page 68 Exercise 21

Suppose that (x_n) is a convergent sequence and y_n is such that for any $\epsilon > 0$ there exists M such that $|x_n - y_n| < \epsilon$ for all $n \geq M$. Does it follow that (y_n) is convergent?

Let $\{x_n\}$ be a convergent sequence with $\lim_{n \rightarrow \infty} x_n = x$

$\Rightarrow \epsilon > 0 \exists N_1 \in \mathbb{N}$ such that $|x_n - x| < \epsilon \forall n \geq N_1$

$\frac{\epsilon}{2} > 0 \Rightarrow |x_n - x| < \frac{\epsilon}{2} \forall n \geq N_1$

$\Rightarrow x - \frac{\epsilon}{2} < x_n < x + \frac{\epsilon}{2}$

From the question we have $\{x_n - y_n\}$ a convergent sequence with $\lim_{n \rightarrow \infty} x_n - y_n = 0$

$\Rightarrow \epsilon > 0 \exists N_2 \in \mathbb{N}$ such that $|x_n - y_n| < \epsilon \forall n \geq N_2$

$\frac{\epsilon}{2} > 0 \Rightarrow |x_n - y_n| < \frac{\epsilon}{2} \forall n \geq N_2$

$\Rightarrow -\frac{\epsilon}{2} < x_n - y_n < \frac{\epsilon}{2}$

Then multiplying the first result by -1 we get, $\Rightarrow -(x + \frac{\epsilon}{2}) < -x_n < -(x - \frac{\epsilon}{2})$

For $n \geq \max\{N_1, N_2\}$, the last two inequalities hold, adding these we get $-x - \epsilon < -y_n < -x + \epsilon$

$\Rightarrow -\epsilon < -y_n + x < \epsilon$

$\Rightarrow |y_n - x| < \epsilon$

$\Rightarrow y_n$ convergent with $\lim_{n \rightarrow \infty} y_n = x$

BS - Page 74 Exercise 7

Let $x_1 := a > 0$ and $x_{n+1} := x_n + \frac{1}{x_n}$ for $n \in \mathbb{N}$. Determine if (x_n) converges or diverges.

Assume for the sake of contradiction that $\{x_n\}$ is convergent with $\lim_{n \rightarrow \infty} x_n = L$

Since the limit of a sequence does not depend on the start point of the sequence, we have

$$\begin{aligned}\lim_{n \rightarrow \infty} x_{n+1} &= \lim_{n \rightarrow \infty} x_n \\ \lim_{n \rightarrow \infty} \left(x_n + \frac{1}{x_n}\right) &= \lim_{n \rightarrow \infty} x_n \\ \lim_{n \rightarrow \infty} x_n + \frac{1}{\lim_{n \rightarrow \infty} x_n} &= \lim_{n \rightarrow \infty} x_n \\ L + \frac{1}{L} &= L \\ \frac{1}{L} &= 0\end{aligned}$$

Thus is a contradiction, thus $\{x_n\}$ is divergent

Note: We know $L \neq 0$ since $a > 0$ and $\{x_n\}$ is monotone increasing,

This is true since $x_n > 0 \forall n$, which we can prove by induction

Base case: $a > 0$

Induction step: Assume $x_k > 0$, thus $x_{k+1} = x_k + \frac{1}{x_k} > 0$,

which implies $x_{n+1} - x_n = \frac{1}{x_n} > 0$ and thus $\{x_n\}$ is monotone increasing